CIRC HQ STUDENT HANDOUT | GRADE 5

Human Program

Goal: Create and debug an exact sequence of commands that guides a human robot through a task.

35 minutes | 8 steps | Computer Science and Programming | Playbook A meeting 14, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Show the original bug and the exact command revision that fixed it.

You will use

- Command cards
- Grid mats
- Safe classroom objects
- Pencils
- Clipboards
- Short-task cards

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one short task card.
- Place the grid mat on the floor.

2 Step 2: Robot safety — 3 min

- Move only at walking speed.
- Stop at every obstacle.
- Stop when anyone says stop.

3 Step 3: Write commands — 5 min

- Write one command per line.
- Use only approved command words.

4 Step 4: Run the program — 7 min

- Give the commands to the human robot.
- Watch each command run exactly.
- Mark the first wrong result.

5 Step 5: Debug one line — 6 min

- Rewrite the first wrong command.
- Run the complete program again.

6 Step 6: Switch roles — 5 min

- Choose a new human robot.
- Run the corrected program once.

7 Step 7: Reset the grid — 3 min

- Stack the command cards.
- Roll the grid mat.

8 Step 8: Exit check — 3 min

- Copy the corrected command.
- Name the result it changed.

What to turn in

The first bug and the command change that fixed it.

No computer? Another way to do it

This lesson is already device free.

Finished early? Next challenge

Loop Finder: Replace a repeated command sequence with one clear repeat instruction.

Stay safe

Human robots move at walking speed and stop immediately at any obstacle or stop command.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.